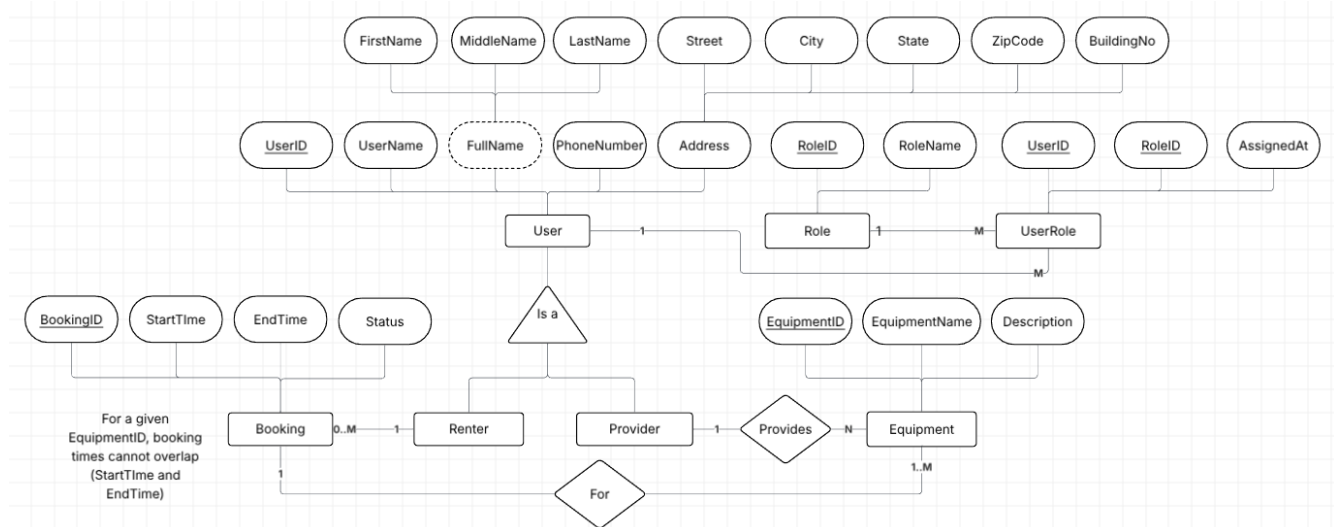


Rustico Dela Cruz III
Professor Daphne Chen
CMPE172 Sec 02
1 March 2026

ERD:



Relationship Schema:

User(UserID, UserName, FirstName, MiddleName, LastName, PhoneNumber, Address)

UserRole(UserID FK → User(UserID), RoleID FK → Role(RoleID), AssignedAt)

Role(RoleID, RoleName)

Renter(UserID, User(UserID))

Provider(UserID, User(UserID))

Equipment(EquipmentID, ProviderID → Provider(UserID), EquipmentName, Description)

Booking(BookingID, RenterID FK → Renter(UserID), EquipmentID FK → Equipment(EquipmentID), StartTime, EndTime, Status)

Explanation:

I want users to be able to be both renters and providers. This is done by having the “Renter” and “Provider” user types be an “is a” relationship, with differentiation with a Role table. This is implemented by having the separated Role table with a RoleID and human-readable RoleName. Renters and Providers of the same user need to be separated as individual users conceptually, so a unique UserID is made for the renter and provider tables while a foreign key from the user table is used.

Equipment and bookings are straightforward:

- start and end times to track bookings and implement booking overlap detection
- status is for a reserved/cancelled attribute
- connected Provider and Renter IDs are foreign key attributes.

I want equipment to be universal, as in this is a loaning system that can hold a numerous amount of items and item types, differentiated by EquipmentIDs and distinguished by the title and description input by Providers.

